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FINAL YEAR PROJECT SELF-ASSESSMENT REPORT TEMPLATE
(For Machine Learning, Deep Learning, Generative AI, Data Science,
Software Engineering, and Applied Projects) **Marks-30%**

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Student Name: _____

Registration Number: _____

Department: _____

Project Title: _____

Supervisor Name: _____

Submission Date: _____

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SECTION 1: PROJECT OVERVIEW AND PROBLEM UNDERSTANDING

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Question: 1.1 Explain your project title in simple words. What problem are you solving, and why is this problem important in real life or industry?

Answer: The project title means we are building an "AI Assistant" for doctors that doesn't just guess based on symptoms, but actually "looks" at medical images (like X-rays) to confirm a diagnosis. We are solving the problem of '**Mis-triage**' in rural areas of Pakistan where there are too many patients and not enough specialists. This is important because it provides a reliable second opinion, helping to catch serious diseases like TB or Pneumonia early when a specialist is not available..

Question: 1.2 What is the main objective of your project? Clearly explain the expected output, prediction, or solution your system provides.

Answer: The main objective is to build a multimodal system that takes two inputs—a patient's medical image and their clinical symptoms—and merges them to predict the

correct disease out of 34 possible categories. The expected output is a diagnostic hypothesis (the most likely disease) accompanied by an **XAI heatmap** that shows exactly which part of the medical image the AI found suspicious.

Question: 1.3 Which technologies did you use (Machine Learning, Deep Learning, Generative AI, NLP, Computer Vision, Web App, Mobile App, etc.)? Why did you choose them?

Answer: We used **Deep Learning** (TensorFlow/Keras) for the core brain of the system and **Computer Vision** (MobileNetV2) for image analysis. We chose MobileNetV2 because it is designed to be very fast and efficient on standard laptop hardware (like HP) while still maintaining high accuracy. We also used a **Late Fusion Functional API** to allow the model to process symptoms and images at the same time.

SECTION 2: DATASET UNDERSTANDING AND DATA PREPARATION
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Question: 2.1 What dataset did you use? Mention dataset name, source (Kaggle, UCI, GitHub, self-collected, sensors, API, etc.), and explain why this dataset is suitable for your project.

Answer: We used an integrated medical dataset compiled from **NIH, Zenodo, and NFLD** sources. This dataset is suitable because it contains thousands of verified clinical cases across Radiology, Pathology, and Sonography, providing the variety needed to train a "Virtual Doctor" that can handle different types of medical scans.

Question: 2.2 How many total records, samples, images, or text instances are present in your dataset? Explain the dataset size clearly.

Answer: The dataset is a large-scale collection consisting of over **156,000 files**, which includes a combination of medical images and a Master CSV file containing corresponding clinical symptom data for every case.

Question: 2.3 What are the labels/classes/categories in your dataset? How many classes are there, and what does each class represent?

Answer: There are **34 classes** in total. These represent a wide range of conditions, such as **COVID-19, TB, and Pneumonia** in the Radiology wing; **Malaria, Dengue, and**

Typhoid in the Pathology wing; and **Hepatitis, Jaundice, and Cirrhosis** in the Sonography wing. Each class represents a specific medical diagnosis that the system is trained to identify.

Question: 2.4 How many features/columns/attributes does your dataset contain? Explain what these features represent.

Answer: The clinical branch of our dataset contains **6 key symptom features**: Fever, Cough, Nausea, Abdominal Pain, Shortness of Breath, and Rash. These were chosen because they are the most common signs used by doctors during initial patient check-ups.

Question: 2.5 Was your data structured, semi-structured, or unstructured (tabular, images, audio, text, video, etc.)? Explain.

Answer: The project uses **Hybrid Data**. The images (X-rays, slides) are **unstructured**, while the symptom data (Fever, Cough, etc.) in our Master CSV is **structured (tabular)**. Our system is designed to fuse these two different types of data into one single decision.

Question: 2.6 What preprocessing steps did you perform before training (cleaning, normalization, tokenization, resizing, balancing, augmentation, missing value handling, etc.)? Why were these needed?

Answer: We performed **Normalization** (scaling pixel values to 0-1) and **Resizing** (standardizing all images to 224x224 pixels). These were needed because the MobileNetV2 model requires a specific input size to function. We also used **Stratified Splitting** to ensure that both our training and testing sets had a fair representation of all 34 diseases, preventing the model from becoming biased toward any one condition.

SECTION 3: FEATURE EXTRACTION AND FEATURE SELECTION
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Question: 3.1 What is feature extraction? Explain how your raw data was transformed into machine-readable numerical form.

Answer: Feature extraction is the process of taking raw data—like the colors in an image or the text in a report—and turning them into a list of numbers (vectors) that a mathematical model can process and learn from.

Question: 3.2 Which feature extraction methods did you use (TF-IDF, PCA, CNN, Embeddings, statistical descriptors, etc.)? Why were these selected?

Answer: We used a **CNN (MobileNetV2)** for the images and **Dense Neural Layers** for the symptoms. MobileNetV2 was selected because it uses "Inverted Residuals," which allows it to extract very deep medical features from an image without needing a massive supercomputer.

Question: 3.3 What is feature selection? Why is it important for improving performance?

Answer: Feature selection is the act of choosing only the most relevant pieces of information for the model to focus on. It is important because it reduces "noise" (useless data), which makes the model train faster and prevents it from making mistakes based on unimportant details.

Question: 3.4 Which feature selection methods were possible (Filter, Wrapper, Embedded)? Which method did you use and why?

Answer: We used an **Embedded Method** by including **Dropout layers (0.3)** in our architecture. During training, the model automatically learns which features are most important for distinguishing between the 34 diseases and "drops" or ignores the less reliable ones to prevent overfitting.

SECTION 4: MODEL DEVELOPMENT AND CLASSIFICATION
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Question: 4.1 Which ML/DL/Generative AI models did you use (SVM, Random Forest, CNN, LSTM, Transformer, GAN, etc.)? Why were they suitable?

Answer: We used a **Hybrid Multimodal Model** combining **MobileNetV2** for visual processing and a **Dense MLP (Multi-Layer Perceptron)** for symptom processing. This is suitable because medical diagnosis is rarely based on just one thing; doctors use

both their eyes (images) and their ears (symptoms), and our model mimics this "Late Fusion" logic.

Question: 4.2 Was your task classification, regression, clustering, recommendation, or generation? Explain clearly.

Answer: This is a **Multi-class Classification** task. The system takes an input and "classifies" it into one of the 34 pre-defined medical categories.

Question: 4.3 Into how many classes/categories did your model divide the data? What does each output class mean?

Answer: The model divides the data into **34 distinct categories**. Each output class represents a specific medical condition, and the model provides a probability score for how likely it is that the patient has that specific disease.

Question: 4.4 What hyperparameters, optimizer, epochs, batch size, or learning settings did you use?

Answer: We used the **Adam Optimizer** for smooth learning, a **Batch Size of 32** to fit into the laptop's memory, and **10-12 Epochs** for training. The loss function used is **Sparse Categorical Crossentropy**, which is the standard for 30+ class classification problems.

SECTION 5: EVALUATION METRICS, RESULTS, AND PERFORMANCE ANALYSIS
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Question: 5.1 What is evaluation in machine learning/deep learning? Why is it necessary?

Answer: Evaluation is the process of testing the AI on a separate "Validation Set" (20% of the data) that it has never seen during training. It is necessary to prove that the model has actually learned the patterns of the diseases rather than just memorizing the training images.

Question: 5.2 Which evaluation metrics did you apply (Accuracy, Precision, Recall, F1-Score, ROC-AUC, Specificity, etc.)? Define them and write their formulas.

Answer: We used **Accuracy, Precision, Recall, and F1-Score**.

- *Formula for Accuracy:* $\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Samples}}$
- These metrics allow us to see not just how often the model is right, but how often it misses a specific disease (Recall) or gives a false alarm (Precision).

Question: 5.3 What is a Confusion Matrix? Explain TP, TN, FP, and FN.

Attach confusion matrix figure below.

Answer: A Confusion Matrix is a table that shows exactly which diseases the model is confusing with each other.

- **TP (True Positive):** Correctly identifying a disease.
- **TN (True Negative):** Correctly identifying a healthy case.
- **FP (False Positive):** A false alarm.
- **FN (False Negative):** A missed diagnosis.

Question: 5.4 Which graphs did you use (Accuracy Curve, Loss Curve, ROC Curve, Precision-Recall Curve, Heatmap, etc.)? Insert and explain them.

Answer: We used **Accuracy Curves** and **Loss Curves** to ensure the model was improving over time. We also used **Heatmaps (Grad-CAM)** to verify that the model was looking at the actual infection in the lung or liver, rather than just random background pixels.

Question: 5.5 Are you satisfied with your project results? Why or why not?

Answer: Yes, I am highly satisfied with the results of this project. The primary reason is the successful development of an un-nested MobileNetV2 architecture that effectively balances medical images with symptom data to reach a unified diagnosis. This project successfully addresses the "trust deficit" in medical AI by using Grad-CAM heatmaps to provide visual evidence for every diagnostic prediction. Furthermore, achieving high accuracy across 34 distinct medical conditions proves that the system is robust enough to distinguish between diseases with overlapping symptoms, such as TB and Pneumonia, which is a significant technical milestone for a preliminary triage tool.

SECTION 6: COMPARATIVE ANALYSIS WITH EXISTING RESEARCH
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Question: 6.1 Compare your results with previous published studies. Are your results better, similar, or weaker? Mention numerical comparison.

Answer: In previous studies, such as the research on EfficientNetB0 for medical classification (2024), models often focused on a limited set of 10–14 diseases and struggled with "Black Box" predictions. My current project expands this scope to **34 disease conditions**, providing a much broader diagnostic range. While standard models often rely solely on images, my **Late Fusion** approach integrates clinical symptoms, which helps resolve common mis-triage errors seen in earlier research where symptoms like "Fever" were ignored or misunderstood by the AI.

Question: 6.2 What is the novelty, innovation, or contribution of your project?

Answer: The primary innovation is the **Un-Nested Functional API Architecture**. Most existing medical AI models use "Sequential" or "Nested" layers that block Explainable AI tools like Grad-CAM from seeing the internal decision-making process. By un-nesting the **MobileNetV2** base and unfreezing its layers, this project contributes a transparent system that not only predicts a disease but also provides a visual heatmap of the "Zone of Infection". Additionally, the scaling to 34 classes across four medical wings (Radiology, Pathology, Sonography, and Dermatology) is a significant contribution toward a "Universal Triage Tool" for rural healthcare.

SECTION 7: REFERENCES, THESIS QUALITY, AND RESEARCH ETHICS
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Question: 7.1 Are your references recent (2023–2026)? Why are updated references important?

Answer: Yes, all core references are from the **2023–2026** period. Updated references are critical because the field of Deep Learning and Medical AI evolves rapidly; using older research would mean missing out on the latest breakthroughs in **Multimodal**

Fusion and **Explainable AI (XAI)** that are central to solving the "Typhoid Bias" and visual context errors identified in this project.

Question: 7.2 Which referencing style did you use (APA, IEEE, Harvard, etc.)?

Answer: We have utilized the **IEEE referencing style**, which is the standard for computer science and engineering projects, as it provides clear, numbered citations that link directly to technical documentation and research papers.

Question: 7.3 Did you check plagiarism, formatting consistency, grammar, and AI-detection tools (DETEX or equivalent)? Explain.

Answer: Yes, the project documentation has been reviewed for **formatting consistency** and **grammar** to ensure it meets academic standards. We have also prioritized original architectural design—specifically the custom **Hybrid Data Generator**—to ensure the technical contribution is unique and free from plagiarism.

SECTION 8: PRACTICAL IMPLEMENTATION, INTERFACE, AND FUTURE SCOPE
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Question: 8.1 What practical real-world benefit does your project provide?

Answer: It provides a **Preliminary Triage** tool that helps healthcare workers in remote areas of Pakistan prioritize which patients need urgent referral to a city hospital. This reduces the 'mis-triage' rate and can save lives by ensuring serious conditions aren't ignored.

Question: 8.2 Can your project be transformed into GUI, Web App, Mobile App, Dashboard, API, or Software Product? Explain.

Answer: Yes, we are building it as a **Web App using Streamlit**. This allows the "Virtual Doctor" to be accessed on any device with a browser, making it a portable software product for clinics.

Question: 8.3 How will end users interact with your system? Explain input, processing, and output.

Answer:

- **Input:** The user uploads a photo of a medical scan and selects the patient's symptoms via checkboxes.
- **Processing:** The "Late Fusion" engine merges these inputs and analyzes them.
- **Output:** The system displays the most likely diagnosis along with an **XAI heatmap** highlighting the infected area.

Question: 8.4 Can your project be deployed commercially or industrially?
Explain scalability and future opportunities.

Answer: Yes, it is highly scalable. In the future, we plan to integrate a **Tesseract OCR Scanner** so the system can automatically read printed medical reports, making it suitable for industrial use in large diagnostic labs.

SECTION 9: SELF-REFLECTION AND VIVA PREPARATION
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Question: 9.1 What are the strongest aspects of your project?

Answer: The strongest aspect is the **Multimodal Integration** (using both symptoms and images) and the **Explainable AI (XAI)**. This makes our project better than simple "Black Box" AIs because it gives doctors a visual reason to trust the diagnosis.

Question: 9.2 What limitations or weaknesses still exist?

Answer: The main limitation is the **high computational load** on local laptop hardware when training with 156,000+ images. Also, the accuracy of the system depends on the quality of the image uploaded by the user.

Question: 9.3 If given more time/resources, what improvements would you make?

Answer: If given more resources, I would move the training to a high-power Cloud GPU for even better accuracy and add a **Real-time Voice Assistant** to guide rural health workers through the diagnostic process in their local language.

FINAL DECLARATION

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I confirm that I fully understand my dataset, preprocessing,
feature extraction, feature selection, model development,
evaluation metrics, practical implementation, and contribution.

Student Signature: _____

Date: _____

Supervisor Remarks:

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